Socioeconomic Development and Household Well-being in South Africa: Predicting Household Food Insecurity

Milestone 4 Final Report

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1. Executive summary

This report covers all six CRISP-DM phases (Wirth and Hipp, 2000) for a model that identifies South African households at risk of food insecurity. It concentrates on the decisions that changed the result.

- **Problem.** Programme planners must decide where to send limited food relief. We ask whether household demographic, housing, service-access and resource characteristics can predict that a household ran out of food, which is the food-insecurity indicator used here (18.4% of households).
- **Approach.** Five survey files were integrated into 20,940 households (20,684 used for modelling), cleaned, and used to compare logistic regression and a tuned random forest under the same 5-fold cross-validation. The best model was deployed as an R/Shiny app with validation and monitoring.
- **Findings.** The random forest ranks households better (test ROC AUC 0.840 against 0.819). At the default 0.50 cut-off it finds only 24.6% of food-insecure households; at a threshold of **0.30** it finds **63.3%** at a precision of **48.8%**. Income, spending, perceived income, dwelling quality and province drive the result, as associations and not causes. Recall differs by province: 73% in KwaZulu-Natal, 49% in Gauteng and 35% in Limpopo.
- **Recommendation.** Deploy the random forest as decision support at 0.30, with a person reviewing every flagged household, monitoring of drift and province-level recall, and revalidation each survey year (Section 11).

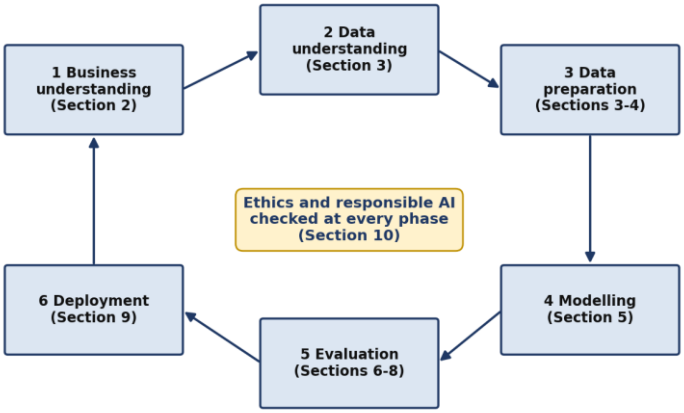


Figure 1. CRISP-DM phases and the sections that cover them.

Where each requirement of the brief is answered

Requirement	Section
Business problem; success criteria	2
Selected data and integration strategy	3
Preprocessing decisions and validation evidence	4
Modelling	5
Evaluation (technical, fairness, business)	6 to 8
Deployment and monitoring	9
Integrated ethics and responsible AI	10
Recommendations linked to results	11
Limitations, assumptions, uncertainty, future work	12 and 13

2. Business understanding

South Africa is highly unequal and household hardship overlaps: poor housing, limited services, low income and hunger. The SDHSA project uses the General Household Survey to target scarce public

resources better than a single indicator such as income can. Milestone 1 set two objectives: (1) a clustering typology of well-being and (2) identifying the drivers that predict household vulnerability, with food insecurity as the outcome. The primary stakeholder is the Department of Planning, Monitoring and Evaluation, with provincial departments, municipalities, development organisations and civil society as the other stakeholders.

The task is supervised **binary classification**. The target FoodInsecure_flag is 1 when the household reported that it ran out of food (RanOutOfFoodCode, file D07), else 0. It is a single-item indicator, not a full scale.

Milestone 1 success criterion	Result	Verdict
At least 3 main and 2 secondary drivers	Five or more consistent drivers (Section 8)	Met
ROC AUC above 0.75; at least 3 variables with $p < 0.05$	AUC 0.840 (CV 0.830); top ten all $p < 0.001$	Met
Importance stable across two approaches	Both models agree on the main drivers	Met
Explainable; presented simply (e.g. decision tree)	Odds ratios and risk bands; no decision tree	Partly met
Specific enough to guide budgets of two provinces	Province-level recall and flag shares only	Partly met
Reproducible and transparent	Notebook, scripts, saved models, repository	Met
Added: recall of at least 60% at the deployed threshold	63.3% at 0.30	Met
Added: recall of at least 40% in every province with 30+ cases	Lowest is Gauteng, 49%	Met

Table 1. Success criteria. Milestone 1 measured ranking quality but not how many at-risk households the tool must find, so the two operating criteria were added.

3. Data selection and integration

Milestone 1 inspected eleven General Household Survey datasets (Statistics South Africa, no date) and selected five that cover well-being without redundancy: D01 population profile (context), D04 dwelling conditions and D05 basic services (core indicators), D07 household well-being (outcome and income) and D09 household resources (enabler). The others were redundant, less relevant or, for D11, shared no household IDs. All records carry survey year 2024. Exploratory Power BI dashboards (distributions, comparisons and correlations, relationships) informed this selection.

D01 is person-level, so it was aggregated to one row per household (size, mean and median age, dependants, dependency ratio) before joining. D04 was the base table, and D05, D07, D09 and the aggregated D01 were added by one-to-one left joins on HouseholdID with the row count checked after each join. Columns present in several sources were renamed by source, and the 22 household-size and 18 income disagreements were flagged, not overwritten.

Stage	Rows	What happened
D01 persons	70,440 to 20,940	25 records with no household ID removed, then aggregated
D04, D07, D09	20,955 to 20,940	Exact duplicates removed (16, 15, 15)
D05	20,940 to 20,925	15 orphan IDs removed; left joins keep those households
Integrated table	20,940	One row per household at every join; 133 columns
Train and test files	20,684	16,548 train and 4,136 test,256 dropped as duplicate

Table 2. Row counts through cleaning and integration.

A data-preparation error. The Milestone 2 modelling step read the 18-digit household IDs as numbers, so 467 different households looked identical and 256 were dropped as "duplicates". The effect is probably small (prevalence 18.4% against 18.3%), but the dropped households were less often food insecure (11%), and the fix is to read IDs as text and rebuild the split (Section 13).

4. Preprocessing decisions and validation evidence

Decision	Reason	Effect
Target = ran out of food, "Insecure" as first level	Clear binary outcome; library treats first level as the event	All metrics describe food-insecure households
All other food-security items removed	They come from the same questions as the target (leakage)	AUC 0.84 reflects real prediction
Placeholder codes (8, 88, 99, runs of 9s) set to missing	They mean "not applicable" or "refused"	e.g. 75% of rent values, 451 income placeholders
Province rebuilt from the numeric code	Up to 24 spellings of 9 provinces	Exactly 9 provinces
Vehicles: 88 to 0, 99 and "Unknown" to missing	88 = no vehicle in 15,217 of 15,232 cases	A true count
Constructed Income_per_capita, services index, age and dependency measures	Affordability, access and composition	Income_per_capita is the top predictor
Nominal codes as categories; 9 and 99 set to missing	A model would read 99 as "more than 14"	No false ordering
Redundant, duplicate and constant columns dropped; rare levels pooled; correlated (above 0.95) columns removed; scaling	Stability; removes near-duplicates such as Age_mean and Age_median	82 predictors remain
Text "NA" read as missing; rent missing-flag; median and mode imputation	77,658 missing cells are stored as text; rent about 75% missing	Correct imputation; missingness kept as information

Table 3. Preprocessing decisions that changed the result.

- **No leakage.** Fixed recodes apply identically to train, test and new data. Everything learned (imputation, levels, scaling, correlation filter) sits in a recipe inside the model workflow, so it is re-fitted within each cross-validation fold. The test set was used once.
- **Validation evidence.** The stratified 80/20 split (seed 381) keeps the target share at 18.4% in both sets; the 4.5 to 1 class ratio was judged moderate and handled by stratification, not resampling. Cross-validation and test results agree (AUC 0.830 and 0.840 for the random forest; 0.819 and 0.819 for logistic regression), and there are no duplicate or overlapping households and no unseen categories in test.

5. Modelling

	Logistic regression	Random forest
Why	Odds ratios are easy to explain to non-technical staff	Captures non-linear effects and interactions
Assumptions	Linear log-odds, no severe multicollinearity, independent households	Few; no linearity or distribution assumption
Parameters	Binomial family, logit link; no tuning	500 trees; mtry 2 to 41 and min_n 2 to 20 on a 5 by 5 grid
Selection	Same 5-fold stratified folds for both	Best cross-validated AUC: mtry 11, min_n 2

Table 4. Candidate models (R tidymodels and ranger; Kuhn and Wickham, 2020; Wright and Ziegler, 2017). The best five random forest settings were within 0.002 AUC.

6. Technical evaluation and threshold

Test set, 0.50 cut-off	Accuracy	Precision	Recall	F1	Specificity	ROC AUC
Random forest	0.840	0.680	0.246	0.361	0.974	0.840
Logistic regression	0.828	0.569	0.272	0.368	0.953	0.819

Table 5. Model comparison on 4,136 test households (cross-validation ROC AUC: 0.830 random forest, 0.819 logistic regression).

Test confusion matrix, 0.50 cut-off	Truth: insecure	Truth: secure	Total
Random forest: predicted insecure	187	88	275
Random forest: predicted secure	573	3,288	3,861
Logistic regression: predicted insecure	207	157	364

Test confusion matrix, 0.50 cut-off	Truth: insecure	Truth: secure	Total
Logistic regression: predicted secure	553	3,219	3,772

Table 6. Confusion matrices at the default cut-off. Both models miss most food-insecure households.

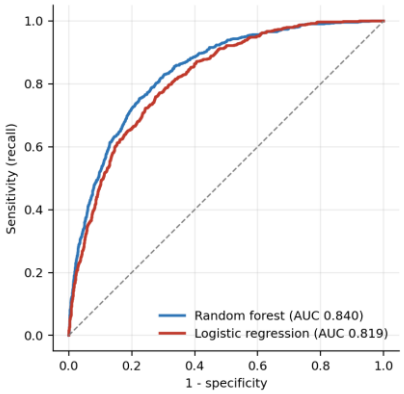


Figure 2. ROC curves, test set.

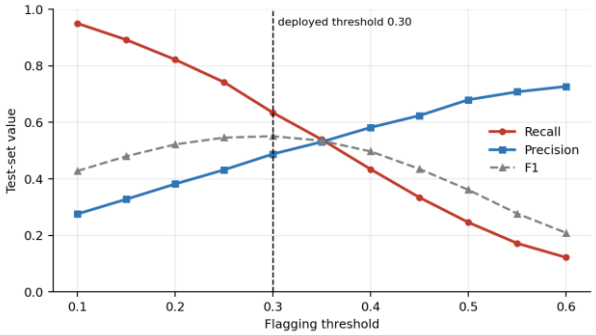


Figure 3. Recall, precision and F1 by threshold (random forest).

Model choice. The random forest has the higher AUC, accuracy, precision (68% against 57%) and specificity; logistic regression has marginally higher recall and F1. We selected the random forest for scoring and use logistic regression to read the direction of effects. **The weakness at 0.50** is that both models find under three in ten food-insecure households, because they rank well but rarely give a probability above 0.50 when only 18% are positive.

Threshold. The cut-off is a business choice about the cost of a missed household against a false alarm. At **0.30** F1 peaks (0.55), recall is 63% and precision 49%; 24% of households are flagged (985 flagged, 481 correctly, 279 vulnerable households still missed). At 0.20 recall is 82% but 40% of households are flagged, probably more than a programme can follow up; at 0.50 precision is 68% but three in four are missed. Probabilities are reasonably calibrated (25% predicted against 22% actual in the 0.2 to 0.3 band).

Threshold	Recall	Precision	F1	Share flagged
0.20	82.2%	38.2%	0.52	39.5%
0.30 (deployed)	**63.4%**	**48.8%**	**0.55**	**23.9%**
0.40	43.4%	58.2%	0.50	13.7%
0.50 (default)	24.6%	68.0%	0.36	6.6%

Table 7. Trade-off at selected thresholds (flagging at or above the threshold, as the app does; counting only scores strictly above 0.30 gives the 63.3% quoted in the text).

7. Fairness across groups

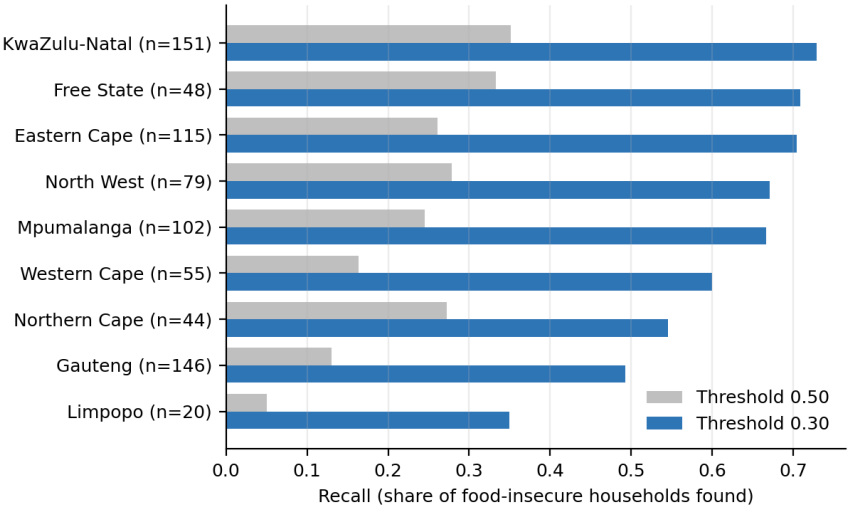


Figure 4. Recall by province at thresholds 0.50 and 0.30 (n = food-insecure test households).

At 0.50 recall ranged from 5% in Limpopo to 35% in KwaZulu-Natal; at 0.30 the range narrows to 35% to 73% but does not close. Limpopo has few positive cases (about 4% food insecure, 20 test cases, interval roughly 18% to 57%), but Gauteng has a large sample (146) and still only 49% recall, so sample size alone does not explain the gap. Population group 1 had recall of 25.5% against 13.0% for group 2 at 0.50, but at 0.30 the gap almost closes (64% against 61%). Groups 3 and 4 have only 1 and 4 cases and cannot be assessed.

Group (food-insecure test households)	n	Recall at 0.50	Recall at 0.30
Head sex code 1	404	25.5%	60.4%
Head sex code 2	356	23.6%	66.9%
Geography type 1	424	20.5%	58.7%
Geography type 2	314	30.6%	69.7%
Geography type 3	22	18.2%	63.6%
Population group 1	701	25.5%	63.9%
Population group 2	54	13.0%	61.1%
Population groups 3 and 4	1 and 4	too few	too few

Table 8. Random forest recall by group on the test set. The 0.30 threshold narrows most gaps; the largest remaining ones are geography type (59% against 70%) and province.

8. Interpretation and business evaluation

Rank	Random forest variable	Logistic predictor	Odds ratio (insecure)	z-value
1	Income_per_capita	ProvinceCode_X9 (Limpopo)	0.38	-17.4
2	Income_clean_D07	ComparativeIncomeCode	1.27	9.9
3	Age_mean	ExpenditureBandCode	0.78	-7.7
4	HeadAge	PayTVAssetCode	1.25	7.3
5	ComparativeIncomeCode	Services_access_index	1.24	6.4
6	SalaryEarnersInHousehold	Income_per_capita	0.29	-6.0
7	ExpenditureBandCode	RoofConditionCode	0.78	-5.8

Table 9. Left: random forest importance ranking. Right: strongest logistic coefficients as odds of food insecurity (the glm models the "secure" class, so values are 1 / odds ratio; numeric predictors are per standard deviation).

Both models point to income, spending, perceived income, dwelling quality and province. A standard deviation more income per capita goes with about 70% lower odds of food insecurity; a worse comparative income rating with about 27% higher odds. These are **associations**: low income goes with

food insecurity, but the data do not show that raising income alone would remove it (Milestone 1 risk R6). Drivers are therefore used to target outreach, never as claimed causes.

What the threshold means in practice. Using the test-set rates (18.4% food insecure) for every 1,000 households screened, a threshold of 0.30 flags about 239 households, finds about 116 of the 184 food-insecure ones and misses about 67, with about 122 false alarms. At 0.50 only about 66 households are flagged, and 139 vulnerable households are missed. The first serves the Milestone 1 goal of finding vulnerable households; the second does not.

Requirement	Evidence on the test set	Verdict
Separates insecure from secure	ROC AUC 0.840 against a target of 0.75	Met
Probabilities work as risk scores	25% predicted against 22% actual (0.2 to 0.3); 77% against 80% above 0.7	Acceptable
Finds the households to reach	Recall 24.6% at 0.50; 63.3% at 0.30	Met at 0.30 only
False alarms tolerable	Precision 48.8% at 0.30; 24% flagged against an 18% base rate	Screening only
Similar across provinces and groups	Recall 73% to 35% by province; small gaps by sex and geography	Partly met
Data current and representative	One survey year (2024); inputs exist only in survey-type data	Revalidate each cycle

Table 10. Deployment readiness of the random forest.

Readiness. The model separates households well (AUC 0.840) and is acceptably calibrated; at 0.30 it finds 63% of vulnerable households, but 49% precision is fine for screening and not for individual decisions, and results are only partly similar across provinces and groups. **Decision: deploy as decision support**, not for automatic decisions about individual households.

9. Deployment and monitoring

Users and decisions. Analysts and planners at the Department of Planning, Monitoring and Evaluation, provincial departments, municipalities and civil society organisations decide which areas and groups to prioritise for relief outreach. **Inputs:** a CSV of households with the Milestone 2 fields and a threshold (default 0.30). **Outputs:** probability, flag and risk band per household; batch counts, flags by province, monitoring status and a downloadable CSV.

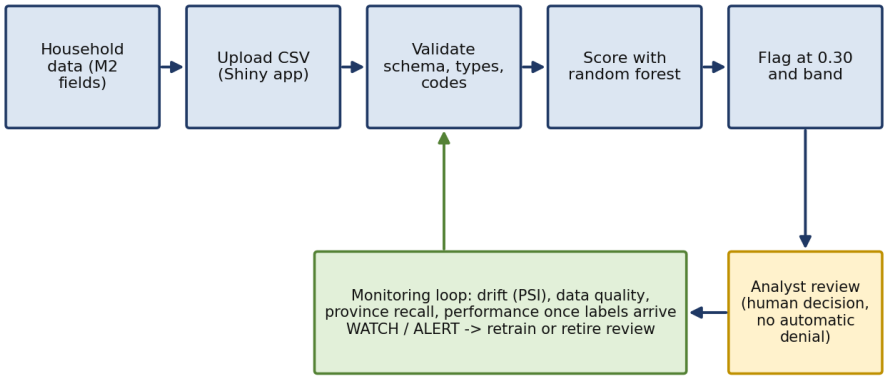


Figure 5. Workflow and monitoring loop. Scores support human decisions and outcomes feed monitoring.

Tool choice. The brief requires an R-based tool, BI.Power BI was used for exploratory dashboard and also final report. Of R/Shiny, a Quarto report, a flexdashboard and a plumber API, **R/Shiny** is the only one giving upload, validation feedback, a threshold control and monitoring in one tool, and it reuses the saved workflow, so app scores match the notebook. Real household data should be hosted inside the organisation.

The app has four tabs: score households (upload, threshold, validation notes, charts, download), a what-if household for explanation only, monitoring, and a model card. **New data** is checked against the trained schema (missing essential columns stop the run), types are converted, fixed cleaning rules apply, columns over 50% missing trigger a warning, and learned preprocessing inside the workflow scores without refitting. Value-range checks are not yet implemented.

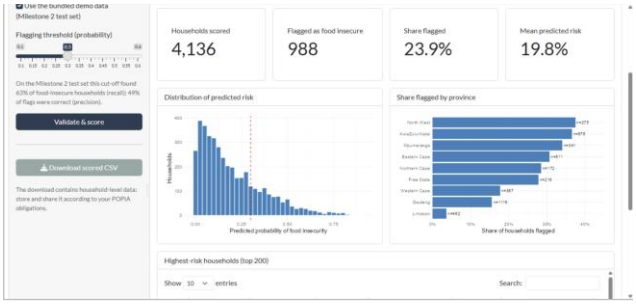


Figure 6. Tab 1, Score households: batch counts, risk histogram and share flagged by province (demo data, threshold 0.30).

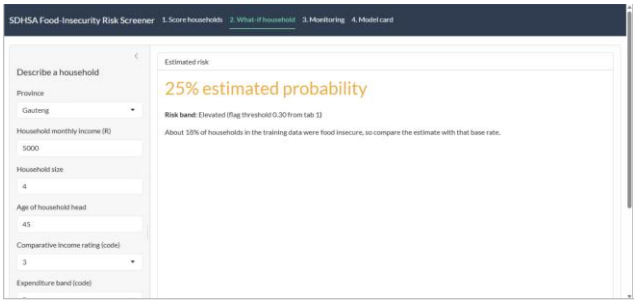


Figure 7. Tab 2, What-if household: estimated probability and risk band for a described household.

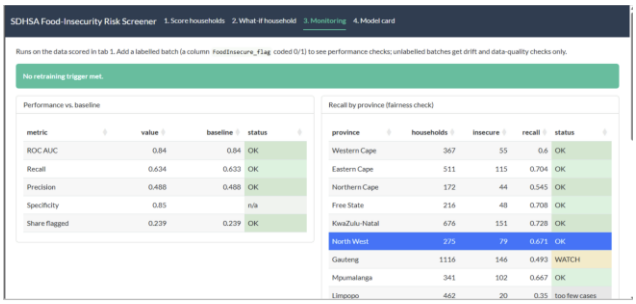


Figure 8. Tab 3, Monitoring: performance against baseline and recall by province.

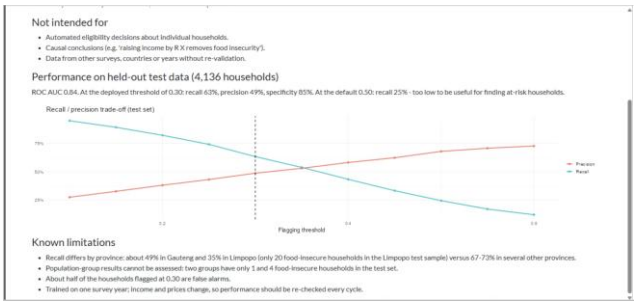


Figure 9. Tab 4, Model card: intended use, test performance, threshold trade-off and limitations.

Criterion	R/Shiny	Quarto report	flexdashboard	plumber API
Interactive upload, threshold, what-if	Yes	Yes	Limited	Needs a client
Usable by non-programmers	High	Medium (read-only)	Medium to high	Low
Shows validation and monitoring live	Yes	Static snapshot	Partly	Data only
Fit for this project	**Best**	Good for periodic reports	Possible	Good later back-end

Table 11. Comparison of R deployment options.

New-data step	What happens	Outcome
1. Schema and types	Compared with trained predictors; numbers sent as text converted	Missing essential columns stop the run
2. Fixed cleaning	Codes 9 and 99 to missing; nominal codes as categories	Same rules as training
3. Quality warning	Columns over 50% missing listed	Scores less reliable
4. Learned preprocessing	Imputation, new categories, dummies, filters, scaling inside the workflow	Nothing refitted on new data
5. Score and check	Probability, flag, band; drift, unseen categories, performance if labelled	OK, WATCH or ALERT

Table 12. Validation and processing of new data. Value-range checks are not yet implemented.

Monitoring measure	Baseline	WATCH	ALERT	Action on ALERT
ROC AUC (needs labels)	0.840	below 0.80	below 0.75	Retrain review
Recall at 0.30 (needs labels)	63%	below 55%	below 50%	Review threshold, retrain
Share flagged	24%	outside 19-29%	outside 15-33%	Look for drift
Province recall (30+ cases)	35-73%	below 50%	below 40%	Fairness review
Feature drift (PSI), 11 drivers	near 0	0.10-0.25	0.25 or more	Three or more: retrain

Monitoring measure	Baseline	WATCH	ALERT	Action on ALERT
Unseen categories; missing share	0%; low	any; over 10%	over 2%; over 20%	Check codes and feed

Table 13. Monitoring limits, our proposed operating rules to be reviewed at each retrain (PSI 0.25 is the usual "major shift" level; Siddiqi, 2006). Run on the saved test predictions, the monitoring reproduces the baseline; a simulated 40% income drop raised the income drift measure to 0.265 (ALERT).

- **Schedule.** Quality and drift checks on every batch; full performance and province review when a new survey year and outcomes are available (about yearly); annual review of threshold, limits and ethics log.
- **Retrain** if AUC falls below 0.75 or recall below 50% on a labelled batch, three or more drivers or the score distribution reach PSI 0.25, survey questions or codes change, or a province falls below the fairness limit. **Retire** if retraining cannot restore AUC above 0.75, the definition of food insecurity changes, key inputs are discontinued, an ethics review finds harm that cannot be removed, or the scores stop being used.
- **Roles and repository.** The BI analyst runs checks and keeps the log, the programme owner confirms the threshold, and a data steward controls access. The repository holds app.R, preprocess.R, monitoring.R, run_monitoring.R, the notebook, saved models and a README

Repository file	Purpose
app.R	Shiny interface and server logic (four tabs)
preprocess.R	Fixed cleaning rules, validate_new_data() and score_households(), shared with the notebook
monitoring.R, run_monitoring.R	PSI, data-quality, performance, fairness and retrain-rule functions; command-line run with a log
Task1.Rmd and notebooks	Modelling pipeline: recipe, cross-validation, tuning, final fit, test evaluation
data/	Saved models (final_rf.rds), train and test data, test predictions, threshold table
README.md, install_packages.R	Setup and run instructions, input requirements

Table 14. Deployment repository.

10. Integrated ethics and responsible AI

Phase	Issue	Response	Remaining risk
Business	Scores used to refuse help; correlation read as cause (R6)	Decision support only; fairness added to criteria	Depends on users
Data	Household survey data; one year; no identifiers needed	No direct identifiers; remove before upload; revalidate yearly	Not representative of other years
Preparation	Placeholders and imputation hide patterns; ID error dropped 256 households	Recoded placeholders, missing-flags; error disclosed	Not yet re-run (Section 13)
Modelling	Head sex and population group are predictors	Neither is in the top ten; used to prioritise, with review	Not tested without them
Evaluation	Unequal recall by province and group	Subgroup metrics, lower threshold, province limit of 40%	Limpopo and Gauteng still under-served
Deployment	Privacy (POPIA), over-reliance, decay	Nothing stored; warning banner; human review; monitoring	No sign-in; relies on hosting

Table 15. Ethical issues across the lifecycle (privacy follows the Protection of Personal Information Act; Republic of South Africa, 2013).

The Milestone 1 ethics log promised to confirm there are no personal identifiers, document missing data, test performance across subgroups and include limitations; all are done, although formal fairness metrics beyond recall are future work and stakeholder engagement is outstanding. Of the Milestone 1 risks, integration failure (R1) did not occur, and overfitting (R8) was controlled by cross-validation and a hold-out set whose results agree. **AI usage:** Claude (Anthropic) helped write and debug the R code, draft report text, troubleshoot metrics, and draft the Shiny app and monitoring functions. The group ran the code, checked figures against real outputs and edited the text; AI-suggested monitoring limits are proposals.

11. Recommendations

1. **Deploy the random forest as decision support at 0.30** (AUC 0.840, recall 63.3%, precision 48.8%; Section 6).
2. **Have the programme owner confirm the threshold**, since it reflects the cost of a missed household against a false alarm.
3. **Keep a person in the loop; never refuse support on a score alone**: about half of flagged households are not food insecure and 279 vulnerable households are still missed.
4. **Check Limpopo and Gauteng first and test province-specific thresholds** (recall 35% and 49% against 73% in KwaZulu-Natal).
5. **Adopt the two operating criteria** (60% recall overall, 40% per province) as formal success criteria.
6. **Run the monitoring plan and revalidate each survey year**, then test the model without head sex and population group.
7. **Describe drivers as associations** and use them to target outreach, not to claim causes.

12. Limitations, assumptions and uncertainty

- **Target.** Food insecurity is one survey item (ran out of food); the code mapping is provisional and 29 "don’t know or refused" answers count as not insecure.
- **Data.** One survey year; survey weights were not used, so results describe the sample, not necessarily the population; 256 households were lost to the ID error; sources disagree on size (22) and income (18); 9 incomes above R500,000 were kept; the services index leaves out electricity.
- **Uncertainty.** The 95% interval around 63% recall is about 60% to 67%; small groups (Limpopo, population groups 3 and 4) are very uncertain; 0.30 and the monitoring limits are judgements.
- **Models.** Odds ratios are per standard deviation and cannot be read as simple group contrasts; the logistic fit warned of near-separation; income is skewed; ExpenditureBandCode is not monotone; impurity importance favours continuous variables and shows no direction; data are cross-sectional.
- **Deployment.** No value-range checks or sign-in; the app was tested on saved test data, not live data.

13. Future work

Read household IDs as text, rebuild the split with all 20,940 households and re-run the models; carry out the Milestone 1 clustering and a decision-tree presentation; check the target coding against the GHS codebook; test province-specific thresholds, class weights or a regularised logistic regression; run a sensitivity test without sex and population group with formal fairness metrics; log-transform income; add range checks, sign-in and an renv lock file; add more survey years and survey weights; and develop budget-level recommendations for provincial departments.

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